SIMATS ENGINEERING

ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1520

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Duration : 60 Minutes

Code of Conduct

I G. Yashmita (192511032) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Yashmita

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Equivalent Steps in VMware Workstation Pro

- Library pane showing multiple VMs = multiple instances/regions.
- VM → Snapshot → Take Snapshot, then Snapshot Manager = backup/versioning demonstration.
- Edit → Virtual Network Editor to show NAT/Host-only/Bridged networks = hybrid or multi-cloud connectivity.
- VM → Manage → Clone = replica/auto-scaling demonstration.

Tip: Paste each screenshot directly under the relevant question in this document (Insert → Picture in Word) before final submission, with a one-line caption explaining what it demonstrates.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded

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1. Comparison of Type-1 and Type-2 Hypervisors:

Type-1 Hypervisor (Bare-Metal):

Type-1 hypervisor runs directly on the physical hardware without requiring a host operating system. Example include VMware ESXI, Microsoft Hyper-V and Xen. Since it interacts directly with hardware, it provides efficient resource allocation, low overhead and better performance.

Type-2 Hypervisor (Hosted):

A type-2 hypervisor runs as an application on top of a host operating system. Examples include Oracle VirtualBox and VMware Workstation. It is easy to install and suitable for development, testing and educational purposes. However it introduces additional overhead because virtual machines depend on the host operating system.

Factor	Type-1 Hypervisor	Type-2 Hypervisor
Performance	High, with lower overhead.	Comparatively lower.
Security	Smaller software stack and reduced host OS exposure.	Depends on the security of the host OS and hypervisor.
Manageability	Supports centralized enterprise management	Easier for individual desktop use
Reliability	Suitable for continuous workloads.	Host OS failures can affect VMs.

A Type-1 hypervisor is the better choice for a cloud datacenter hosting mission-critical applications because it offers better performance, stronger isolation architecture, higher reliability and centralized management capabilities. Regular security patching, access control, monitoring and high-availability configuration should also be implemented to maintain secure and continuous service.

2. How Virtualization Supports Isolation :

1. Virtual Machine Isolation :
Each department can operate inside a separate virtual machine with its own operating system, applications and allocated resources.

2. Network Isolation :
Virtual networks, VLANs, virtual switches and firewall rules can separate finance, healthcare and student traffic.

3. Storage Isolation :
CPU and memory reservations, limits and quotas help prevent one workload from consuming all available resources.

4. Resource Isolation :
Separate virtual disks, encrypted storage volumes and access permissions prevent unauthorized access to confidential information.

Two major Attack surfaces:

1. Hypervisor and Virtualization Management Interface:

The hypervisor manages virtual machines and control access to physical resources.

2. Virtual Networks and Inter-VM communication:

Virtual switches, network interfaces and configuration rules may allow unauthorized communication.

3. VM Host Utilization and Correctness Actions:

The given VM host has CPU utilization of 92%, memory utilization of 88%, storage I/O latency of 35 ms and average VM boot time of 170 seconds.

CPU Utilization:

High CPU usage indicates that multiple VMs are competing for processing resources. The administrator should identify CPU-intensive VMs, optimize workloads and distribute them

across other hosts.

Memory Utilization:

Memory usage of 88.1. indicates limited available RAM. Unnecessary processes should be stopped. VM memory allocations should be optimized and additional RAM can be installed, if required.

Storage I/O Latency:

A latency of 35 ms may indicate slow storage devices or excessive disk requests. Using SSD storage, balancing workloads and optimizing disk operations can improve performance.

VM Boot Time:

The average boot time of 170 seconds may result from resource contention and slow storage. Unnecessary startup services should be disabled and VM startup operations should be optimized.

4. Software Defined Datacenter (SDDC) Architecture:

A software Defined Datacenter (SDDC) is an architecture in which

computing, storage and networking resources are managed through software -

Compute Virtualization:

Hypervisors allow multiple virtual machines to run on a single physical server. Administrators can create VMs quickly, allocate CPU and memory and move workloads between compatible hosts.

Storage Virtualization:

Physical storage resources are combined into logical storage pools. This allows administrators to allocate storage dynamically, manage backups and increase capacity without extensive manual configuration.

Network Virtualization:

Software defined networking enables virtual networks, centralized configuration and security policies.

Automation and Agility:

SDDC uses software tools and APIs to automate resource provisioning and monitoring. This reduces manual work, speeds up application deployment and improves resource utilization.

5. Multi-Cloud Federation and Identity Mismatches:

Multi-cloud federation allows an organization to access services across multiple cloud providers using trusted identities relationships. Identity mismatches occur when providers use different authentication systems.

Causes:

One provider may identify users by email address, while another uses unique identifiers.

Proposed Solution:

1. Centralized Identity Provider:

Use a trusted identity provider to manage user authentication across cloud platforms.

2. Identity Federation:

Implement standards such as SAML or OpenID connect to establish secure trust between the identity provider and cloud services.

3. Role Mapping:

Map user identities and roles consistently across providers using role-

based access control (CRBAC).

4. Security Controls:
Enforce multi-factor authentication, least-privilege access, short-lived tokens and regular permission reviews.

5. Privacy protection:
Share only necessary identity attributes, encrypt data during transmission and protect sensitive information.

6. Monitoring:
Maintain audit logs and revoke access immediately when users leave the organization or credentials are compromised.

A centralized identity provider, standard-based federation, consistent role mapping and strong security controls can resolve identity mismatches and provide secure, privacy-preserving access across multiple cloud environments.